

Evans Syndrome: A Comprehensive Disease Characteristics Report

Disease: Evans syndrome **MONDO ID:** MONDO:0016030 · **Orphanet:** ORPHA:1959 · **Category:** Autoimmune cytopenia / immune dysregulation **Report type:** Literature-based disease knowledge-base entry (no primary datasets provided; all content derived from aggregated disease-level literature and cohort studies).

Summary

Evans syndrome (ES) is a rare autoimmune disorder defined by the **concurrent or sequential occurrence of autoimmune hemolytic anemia (AIHA)** — proven by a positive direct antiglobulin (Coombs) test with hemolysis — **and immune thrombocytopenia (ITP)**, sometimes accompanied by autoimmune neutropenia. It is a diagnosis of exclusion, established with a complete blood count, blood smear, and Coombs test after secondary causes have been ruled out. The final common effector mechanism is production of **warm IgG autoantibodies** against erythrocyte antigens and anti-platelet glycoprotein antibodies, leading to **Fc-receptor- and complement-mediated destruction of red cells and platelets, predominantly in the spleen.**

The central insight consolidated across this investigation is that Evans syndrome is best understood **not as an isolated blood disorder but as a phenotype of systemic immune dysregulation.** In children especially, ES is frequently the presenting manifestation of an underlying **inborn error of immunity (IEI)** — with monogenic or immune-dysregulation causes identified in roughly 50–60% of pediatric autoimmune-cytopenia cohorts (ALPS/FAS pathway, CTLA4, LRBA, PIK3CD/APDS, NFKB1, SASH3, partial DiGeorge syndrome). It may also be secondary to systemic lupus erythematosus (SLE), lymphoproliferative disease, or common variable immunodeficiency. ANA positivity in childhood ES is a strong predictor of progression to SLE.

The clinical course is **chronic and relapsing with substantial morbidity and mortality.** Nationwide registry data give an adult incidence approaching ~1.8 per million person-years and a median survival of ~7 years overall (dramatically worse — ~1.7 years — for secondary

ES). Pediatric-onset disease carries ~16% mortality at 15 years, most often from infection or bleeding, with a very high treatment burden. Management escalates from corticosteroids/IVIG (first-line) to rituximab and sirolimus (steroid-sparing second-line), with genotype-directed targeted immunotherapy (abatacept, leniolisib, JAK inhibitors) increasingly important. This report details all 15 disease-characteristic domains, flagging where evidence is robust versus where it is sparse or not applicable.

Key Findings

F001 — Definition: coexistence of AIHA and ITP

Evans syndrome is consistently defined across cohorts as the **simultaneous or sequential association of AIHA (positive direct antiglobulin/Coombs test with hemolysis) and ITP**. The two events may occur at the same time or one may follow the other. Diagnosis relies on a full blood count/film and a Coombs test, and is a diagnosis of exclusion after secondary causes are ruled out. In the French OBS'CEREVANCE pediatric AIHA cohort (n=265), Evans syndrome was present in **37%** of childhood AIHA cases.

"There is a coexistence of Immune thrombocytopenia (ITP) with Autoimmune haemolytic anaemia (AIHA) and both of these events may occur simultaneously or one follows the other." — [PMID: 31983745](#) "Evans' syndrome was diagnosed in 37% of cases." — [PMID: 21228033](#)

F002 — Frequently the presenting sign of an inborn error of immunity (IEI)

In a Tampa Bay prospective autoimmune cytopenia (AIC) cohort (n=104), **51% showed evidence of IEI and 26% had monogenic disorders**; IEI prevalence was highest in Evans syndrome (61.5%) and AIHA (62.5%). The most common monogenic causes were **partial DiGeorge syndrome (pDGS)**, followed by variants in **NFKB1, CTLA4, and FAS**. A germline **SASH3** nonsense mutation (c.862C>T; p.Arg288Ter) has been identified as a specific monogenic cause. By contrast, an adult chronic ITP/Evans cohort (n=44) screened with an NGS panel of >370 IEI genes found **no fully penetrant IEI**, though 18.2% carried heterozygous pathogenic IEI variants — establishing that the diagnostic yield of IEI testing is far higher in pediatric than adult-onset disease.

"Among 104 AIC patients, 53 (51%) showed evidence of IEI, including 27 (26%) with monogenic disorders-most commonly partial DiGeorge syndrome (pDGS), followed by variants in NFKB1, CTLA4, and FAS." — PMID: 41560547 "No cases of IEI were identified despite a high representation of subjects with a personal history of autoimmunity" — PMID: 37792884 "identified a germline mutation in SASH3 (c.862C>T;p.Arg288Ter), indicating a recently identified IEI" — PMID: 37646304

F003 — Chronic, relapsing, multisystem course with ~16% pediatric mortality at 15 years

In the French OBS'CEREVANCE cohort of 151 pediatric-onset ES patients with >5 years follow-up (median 11.3 years): at 10 years, ITP and AIHA were in sustained complete remission in 54.5% and 78.4% respectively; by age 20, **74% had ≥ 1 clinical immunopathological manifestation** (lymphoproliferation, dermatological, GI/hepatic, pulmonary). **Survival at 15 years was 84% (~16% mortality)**; death occurred at a median age of 18 years, most often from infection. The **number of second-line treatments and severe/recurrent infections were independently associated with mortality**. In an earlier cohort (n=156), 5-year ITP and AIHA relapse-free survival were 25% and 61%, and 69% required ≥ 1 second-line treatment.

"At 10 years, ITP and AIHA were in sustained complete remission in 54.5% and 78.4% of patients, respectively." — PMID: 33440924 "Survival at 15 years after diagnosis was 84%." — PMID: 33440924 "Overall, 69% of children required one or more second-line immune treatments" — PMID: 26484337

F004 — Treatment: corticosteroids/IVIG first-line; rituximab and sirolimus effective second-line

Standard first-line therapy is **corticosteroids $\pm$ IVIG**, but relapse on taper is common. **Rituximab** (anti-CD20): in a prospective French pediatric AIHA/Evans cohort (n=61), 75% responded and rituximab allowed steroid withdrawal in 72%; 6-year relapse-free survival was 48% (higher in isolated AIHA than in ES, $P < 0.05$). **Sirolimus** (mTOR inhibitor): in a multicenter prospective trial of 30 refractory AICs, all 12 ALPS children achieved durable complete response and most patients with Evans syndrome/CVID/SLE responded. **Genotype-directed**

targeted agents are emerging (abatacept for CTLA4/LRBA; sirolimus/leniolisib for ALPS/APDS; ruxolitinib/baricitinib). **Splenectomy** is effective but its benefit is diminished by immunopathological manifestations and carries infection/thrombosis risk.

"Forty-six patients responded (75%) and the 6-year relapse-free survival (RFS) was 48%."
— PMID: 28444729 "sirolimus led to CR and durable responses in a majority of children with refractory multilineage autoimmune cytopenias" — PMID: 26504182

F005 — ANA-positive childhood ES is a strong risk factor for progression to SLE

In the OBS'CEREVANCE cohort, ANA were positive in 20% (355/1803) of children with AIC; 22% of ANA-positive patients developed SLE at a median age of 14.5 years. **Progression to SLE occurred in 45% of ANA-positive Evans syndrome patients** (vs 20% chronic ITP, 19% AIHA); no ANA-negative patient developed SLE. Independent risk factors were **age >10 years at AIC diagnosis (RR 3.67, 95% CI 1.18–11.4, P=.024)** and **ANA titer >1/160 (RR 5.28, 95% CI 1.20–23.17, P=.027)**.

"20% of chronic immune thrombocytopenic purpura, 19% of autoimmune hemolytic anemia, and 45% of Evans syndrome" — PMID: 38227934

F006 — Very rare, rising incidence; secondary ES has markedly worse survival

Danish nationwide registry data. **Adults (n=242, 1977–2017)**: mean age at diagnosis 58.5 years, 51.2% women, 27.3% secondary; incidence rose to **1.8 per million person-years** and prevalence to **21.3 per million persons** by 2016. Median survival was **7.2 years overall (primary 10.9 years; secondary only 1.7 years; secondary 5-year survival 38%)**; leading causes of death were bleeding, infections, and hematological cancer. **Children <13 years (n=21)**: incidence 0.5–1.2 per million person-years; prevalence rose from 6.7 (1990) to 19.3 (2015) per million; hazard ratio for death was **22-fold** higher than matched general-population children.

"The annual Evans syndrome incidence and prevalence rose significantly during the study period, to 1.8 per million person-years and 21.3 per million persons, respectively, in 2016." — [PMID: 31292991](#) "The median survival with Evans syndrome was 7.2 years (primary Evans syndrome: 10.9 years; secondary Evans syndrome: 1.7 years)." — [PMID: 31292991](#) "Hazard ratio for death was 22 fold higher for children with ES compared to matched children from general population" — [PMID: 32271826](#)

F007 — Mechanism: defective lymphocyte apoptosis / broken tolerance → autoantibody-mediated cytopenias

A large subset of ES — particularly **ALPS** — results from defective **FAS-mediated extrinsic apoptosis** of lymphocytes (germline/somatic **FAS**, **FASLG**, **CASP10**), leading to lymphoproliferation, expansion of **TCRαβ+ CD4–CD8– double-negative T (DNT) cells**, and autoimmune cytopenias. ALPS biomarkers include elevated DNT cells and elevated **soluble FAS ligand (sFASL)**. Other monogenic causes disrupt tolerance at distinct nodes: **CTLA4/LRBA** (impaired Treg checkpoint), **PIK3CD** (APDS, PI3K-δ hyperactivation), **NFKB1** (haploinsufficiency), and **SASH3** (lymphocyte adaptor; germinal-center hypoplasia). The final common step is warm IgG anti-erythrocyte and anti-platelet-glycoprotein autoantibody production causing Fc-receptor/complement-mediated splenic destruction.

"Autoimmune lymphoproliferative syndrome (ALPS) is a disorder of disrupted lymphocyte homeostasis, resulting from mutations in the Fas apoptotic pathway." — [PMID: 22157362](#) "sFASL level can efficiently discriminate patients with ALPS when using the appropriate thresholds" — [PMID: 38700373](#) "LRBA protein deficiency was shown to be responsible for different types of inborn errors of immunity, such as common variable immunodeficiency (CVID) and autoimmune lymphoproliferative syndrome (ALPS)" — [PMID: 31432443](#)

F008 — Clinical phenotype and diagnostic laboratory profile

The phenotype combines features of both cytopenias. **AIHA component** (HP:0004808): pallor (HP:0000980), fatigue (HP:0012378), jaundice (HP:0000952), dark urine, splenomegaly (HP:0001744); labs show positive DAT (warm IgG±C3d — **74% IgG/IgG+C3d** in the pediatric cohort), reticulocytosis, elevated LDH and indirect bilirubin, low haptoglobin. **ITP component** (HP:0001973): petechiae (HP:0000967), purpura/bruising (HP:0000979), mucosal/GI bleeding

(HP:0011897). When ES reflects immune dysregulation, organomegaly is prominent — in LRBA deficiency **splenomegaly occurred in 93.3% (14/15)**. Some patients also have neutropenia (HP:0001875), hypogammaglobulinemia (HP:0004313), and recurrent infections (HP:0002719).

"In 74% of cases the direct antiglobulin test was IgG/IgG+C3d." — [PMID: 21228033](#)
 "Splenomegaly was seen in 93.3% (14/15) of the patients on admission." — [PMID: 31432443](#)

F009 — Natural disease in dogs; FAS-pathway (Fas^{lpr}) mouse model

Natural disease: ES (concurrent immune-mediated hemolytic anemia + immune-mediated thrombocytopenia) is a recognized spontaneous entity in the domestic dog (*Canis lupus familiaris*, NCBI:txid9615), reported across breeds including Rottweiler, Miniature Schnauzer, and Dachshund; canine cases are DAT-positive, glucocorticoid/immunosuppressant-responsive, and can be complicated by thrombosis and opportunistic infection — paralleling human disease. **Model organism:** the FAS pathway underlying ALPS-type human ES is modeled by the **MRL/lpr (Fas^{lpr}) and gld (Faslg)** mouse, which develop lymphoproliferation, TCRαβ⁺ DNT-cell accumulation, autoantibodies, and autoimmune cytopenias; leniolisib reduced lymphoproliferation in murine ALPS.

"presumptively diagnosed with Evans' syndrome (ES)" — [PMID: 19411652](#) "Leniolisib reduced lymphoproliferative disease in murine autoimmune lymphoproliferative syndrome" — [PMID: 41608120](#)

The 15-Section Disease Characteristics Report

1. Disease Information

Overview. Evans syndrome is a rare, chronic autoimmune disorder characterized by the combination of AIHA and ITP (± autoimmune neutropenia), occurring simultaneously or sequentially (F001). It is a diagnosis of exclusion made after ruling out secondary causes (SLE, lymphoproliferative disease, IEL, drugs, infection).

Key identifiers: | Resource | Identifier | |---|---| | MONDO | MONDO:0016030 | | Orphanet | ORPHA:1959 | | ICD-10 | D69.3 (ITP component) / D59.1 (AIHA component) — no single dedicated code | | ICD-11 | 3B51.0 / 3A20 (component-based) | | MeSH | Anemia, Hemolytic, Autoimmune; Thrombocytopenia (no unique ES MeSH heading) | | OMIM | No single OMIM entry; monogenic causes have their own (e.g., ALPS 601859; CTLA4 haploinsufficiency 616100; APDS 615513) |

Synonyms: Evans-Fisher syndrome; autoimmune hemolytic anemia with immune thrombocytopenia; combined autoimmune hemolytic anemia and thrombocytopenia.

Information source. Evidence in this report derives from aggregated disease-level resources: national prospective cohorts (French OBS'CEREVANCE), nationwide registries (Danish), and case series/reports — not from a single EHR extract.

2. Etiology

Causal factors. ES is heterogeneous. It may be **primary (idiopathic)** or **secondary** to another disorder. A dominant paradigm is that ES is often the hematologic expression of an **underlying inborn error of immunity** (F002, F007). Documented monogenic/genetic causes: **FAS**, **FASLG**, **CASP10** (ALPS), **CTLA4**, **LRBA** (Treg checkpoint defects), **PIK3CD** (APDS), **NFKB1** (haploinsufficiency), **SASH3**, and **partial DiGeorge syndrome (22q11.2 deletion)**. Secondary causes include SLE, common variable immunodeficiency, and lymphoproliferative disease.

Genetic risk factors. Germline loss-of-function variants in the genes above; somatic FAS variants in ALPS; the 22q11.2 microdeletion (pDGS). ANA positivity marks a susceptibility state for SLE-associated ES (F005).

Environmental risk factors. Age (bimodal: pediatric and older-adult peaks), female sex (adult 51.2% women), family/personal history of autoimmunity. No established toxin, occupational, or dietary cause. Some secondary cases follow infection or lymphoma.

Protective factors. None specifically established. ANA-negativity predicts against SLE progression (F005). No protective genetic alleles are defined for ES.

Gene–environment interactions. In ANA-positive children, age >10 years plus high ANA titer (>1/160) multiplicatively raise SLE risk (F005), illustrating interaction between an autoantibody "environment" and host age/genetic background. Not otherwise well characterized.

3. Phenotypes

Phenotype	Type	HPO term	Frequency / notes
Autoimmune hemolytic anemia	Lab/clinical	HP:0004808	Defining; DAT+ in ~all; 74% IgG/IgG+C3d
Autoimmune thrombocytopenia	Lab/clinical	HP:0001973	Defining
Pallor	Sign	HP:0000980	Common (anemia)
Fatigue	Symptom	HP:0012378	Common
Jaundice	Sign	HP:0000952	Hemolysis
Splenomegaly	Sign	HP:0001744	Very common in immune-dysregulation ES; 93.3% in LRBA deficiency
Petechiae	Sign	HP:0000967	Thrombocytopenic bleeding
Purpura/bruising	Sign	HP:0000979	Common
Mucosal/GI bleeding	Sign	HP:0011897	Variable severity
Lymphadenopathy	Sign	HP:0002716	ALPS-type ES
Neutropenia	Lab	HP:0001875	Subset (triple cytopenia)
Hypogammaglobulinemia	Lab	HP:0004313	IEI-associated
Recurrent infections	Clinical	HP:0002719	IEI-associated; major cause of death

Characteristics. Onset spans neonatal to geriatric; pediatric-onset disease is typically chronic and relapsing (F003), adult disease often chronic. Severity is **variable to severe** (life-threatening anemia or bleeding possible). Progression is **episodic/relapsing-remitting**. By age 20, 74% of pediatric-onset patients have ≥ 1 additional immunopathological manifestation (F003).

Quality-of-life impact. High treatment burden, chronic relapses, transfusion dependence in flares, immunosuppression-related infection risk, and cumulative organ involvement substantially impair QoL; formal EQ-5D/SF-36/PROMIS data specific to ES were not identified (knowledge gap).

4. Genetic / Molecular Information

Causal genes (in monogenic/IEI-associated ES): **FAS** (OMIM 134637), **FASLG**, **CASP10**, **CTLA4**, **LRBA**, **PIK3CD**, **NFKB1**, **SASH3**; chromosomal **22q11.2 deletion** (partial DiGeorge) (F002, F007).

Pathogenic variants. Documented examples include the **SASH3** nonsense variant **c.862C>T (p.Arg288Ter)** (F002). Variant classes span nonsense/frameshift (loss of function; NFKB1, LRBA, SASH3), missense (gain of function in PIK3CD/APDS), and structural (22q11.2 deletion). ALPS may involve **somatic** FAS variants restricted to DNT cells as well as germline variants. Functional consequences: **loss of function** (FAS apoptosis, LRBA/CTLA4 checkpoint, NFKB1), **gain of function** (PIK3CD/PI3K- δ hyperactivation).

Modifier genes. Not formally established for ES; disease expression is modified by the specific IEI genotype and by secondary triggers (SLE, lymphoma).

Epigenetic information. No ES-specific DNA-methylation or histone-modification signature was identified in the literature reviewed (knowledge gap).

Chromosomal abnormalities. 22q11.2 deletion (partial DiGeorge) is the single most common monogenic-level cause in one large pediatric AIC cohort (F002).

5. Environmental Information

Environmental factors. No established chemical, radiation, or occupational cause. **Infectious agents** may act as triggers in secondary ES, and infections are a leading complication/cause of death, but no single pathogen is causal. **Lifestyle factors** are not established causes; comorbidities such as diabetes and heavy smoking appear in case reports of thrombotic complications but are not disease causes. ES itself is autoimmune, not infectious or transmissible.

6. Mechanism / Pathophysiology

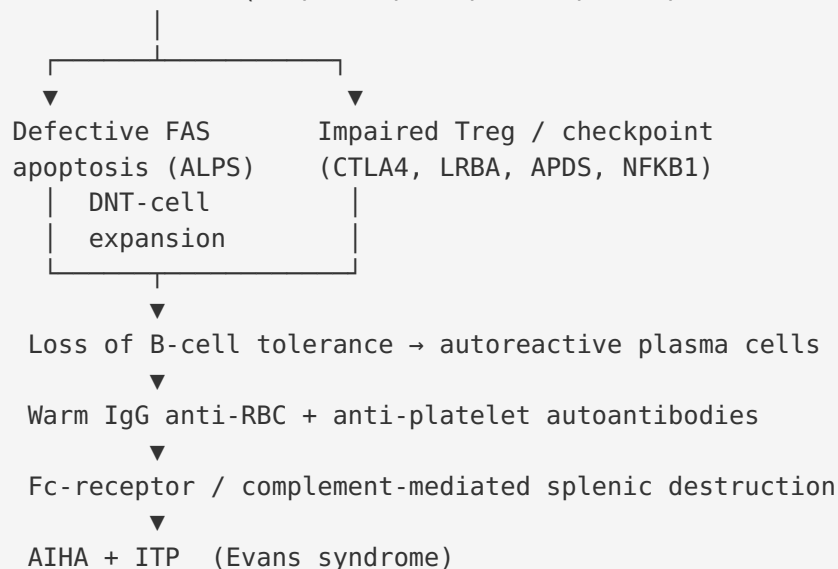
Ordered causal chain (initiating lesion → clinical manifestation):

1. A germline (or somatic) genetic lesion in a tolerance/apoptosis gene — e.g., **FAS/FASLG/CASP10, CTLA4, LRBA, PIK3CD, NFKB1, SASH3** — **leads to** defective lymphocyte homeostasis or a broken immune checkpoint (upstream; demonstrated for these IEIs). *In primary/idiopathic ES without an identified gene, this step is inferred.*
2. **Branch A (ALPS-type):** defective FAS-mediated extrinsic apoptosis **results in** failure to delete autoreactive lymphocytes → lymphoproliferation and **accumulation of TCR $\alpha\beta$ ⁺**

CD4–CD8– double-negative T (DNT) cells (demonstrated; DNT cells and elevated soluble FASL are diagnostic biomarkers).

3. **Branch B (checkpoint-type):** CTLA4/LRBA loss, PI3K- δ hyperactivation (APDS), or NFKB1 haploinsufficiency **results in** impaired regulatory T-cell function and dysregulated B/T-cell activation.
4. Both branches **converge on** loss of B-cell tolerance and **results in** activation of autoreactive B cells / plasma cells.
5. This **leads to** production of **warm IgG autoantibodies** against red-cell antigens and anti-platelet glycoprotein (e.g., GPIIb/IIIa) antibodies (demonstrated effector step).
6. Autoantibody opsonization **results in Fc-receptor– and complement-mediated phagocytosis/destruction of erythrocytes and platelets, predominantly by splenic macrophages** (downstream effector).
7. This **leads to** the clinical manifestations: hemolytic anemia (pallor, jaundice, reticulocytosis, low haptoglobin, high LDH/bilirubin) and thrombocytopenia (petechiae, purpura, bleeding) (F007, F008).

Genetic lesion (FAS/CTLA4/LRBA/PIK3CD/NFKB1/SASH3 or unknown)



Molecular pathways: FAS/FASLG extrinsic apoptosis (caspase-8/10); PI3K-AKT-mTOR (APDS; rationale for sirolimus/leniolisib); CTLA4–CD80/86 costimulation checkpoint; NF- κ B signaling.

Cellular processes: defective apoptosis, lymphoproliferation, breakdown of self-tolerance, complement activation, macrophage phagocytosis. **Immune involvement:** combined autoimmunity + immunodeficiency (recurrent infection). **GO terms:** apoptotic process (GO:0006915), regulation of immune response (GO:0050776), complement activation

(GO:0006956), phagocytosis (GO:0006909). **Cell types (CL):** double-negative T cell, regulatory T cell (CL:0000815), B cell/plasma cell (CL:0000786), macrophage (CL:0000235), erythrocyte (CL:0000232), platelet/thrombocyte (CL:0000233).

7. Anatomical Structures Affected

Organ level. Primary: **spleen** (UBERON:0002106; principal site of destruction and often enlarged), **bone marrow** (UBERON:0002371; compensatory hyperplasia), **blood** (UBERON:0000178). Secondary/associated: **lymph nodes** (UBERON:0000029; lymphoproliferation), **liver** (UBERON:0002107; hepatomegaly, associated autoimmune hepatitis), lungs and GI tract in multisystem ES. Body systems: hematopoietic/immune (primary), with cardiovascular (thrombosis risk), hepatic, pulmonary, and GI involvement in immune-dysregulation subtypes.

Tissue/cell level. Targets: **erythrocytes** (CL:0000232) and **platelets** (CL:0000233); effectors: splenic **macrophages** (CL:0000235), dysregulated **T cells** (including DNT cells) and **B cells/plasma cells**.

Subcellular level. Death-inducing signaling complex at the plasma membrane (FAS/FADD/caspase); autoantibody targets are red-cell membrane proteins and platelet-surface glycoproteins. GO cellular components: plasma membrane (GO:0005886), death-inducing signaling complex (GO:0031264).

Localization. Systemic; splenic destruction is central. Lateralization not applicable (systemic hematologic disease).

8. Temporal Development

Onset. Bimodal — pediatric (childhood) and older-adult (mean 58.5 years in the Danish adult cohort). Onset pattern is typically **subacute to chronic/insidious**, though acute severe hemolytic or bleeding crises occur. The two cytopenias may present simultaneously or years apart (F001).

Progression. **Relapsing-remitting/episodic**, chronic lifelong course (F003). Pediatric 5-year relapse-free survival: ITP 25%, AIHA 61%. At 10 years, sustained CR in 54.5% (ITP) and 78.4% (AIHA). Multisystem immunopathology accrues over time (74% by age 20).

Patterns. Remissions are usually **treatment-induced**; spontaneous durable remission is uncommon in pediatric-onset disease. Critical windows: early diagnosis of an underlying IEI opens the door to genotype-targeted therapy; monitoring for immunopathological manifestations refines splenectomy risk–benefit.

9. Inheritance and Population

Epidemiology (F006). Adult incidence ~**1.8 per million person-years**, prevalence ~**21.3 per million** (Denmark, 2016). Pediatric incidence 0.5–1.2 per million person-years; pediatric prevalence rising (6.7→19.3 per million, 1990→2015). Adult mean age at diagnosis 58.5 years; 51.2% women; 27.3% secondary.

Genetic etiology. Inheritance depends on the underlying IEI: **autosomal dominant** (ALPS/FAS, CTLA4 haploinsufficiency, NFKB1, PIK3CD GOF), **autosomal recessive** (LRBA), **X-linked** (SASH3), or **de novo/structural** (22q11.2 deletion). Penetrance is **incomplete and age-dependent** (e.g., CTLA4 haploinsufficiency). Expressivity is variable. Somatic mosaicism occurs in ALPS (somatic FAS variants). Anticipation, founder effects, and consanguinity are not general features of ES, though consanguinity increases recessive-IEI (LRBA) risk. Most "primary" ES has no identified single gene and behaves as a multifactorial/polygenic autoimmune trait.

Population demographics. No strong ethnic predilection established; adult female predominance is modest (~51%). Geographic distribution is worldwide; rising recorded incidence likely reflects better ascertainment.

10. Diagnostics

Clinical/laboratory tests (F008). CBC with **blood smear** (spherocytes, polychromasia, low platelets); **direct antiglobulin (Coombs) test** (warm IgG ± C3d; 74% IgG/IgG+C3d); reticulocyte count (elevated); **LDH** (elevated), **indirect/unconjugated bilirubin** (elevated), **haptoglobin** (low); platelet count (low). LOINC-codable analytes: hemoglobin, platelet count, reticulocytes, LDH, bilirubin, haptoglobin, DAT.

Biomarkers. For underlying ALPS: elevated **TCRαβ⁺ DNT cells** and **soluble FAS ligand (sFASL)**; also elevated vitamin B12 and IL-10 in ALPS. ANA (SLE risk marker). Immunoglobulin levels (hypogammaglobulinemia in IEI).

Genetic testing. Given the high IEI yield in pediatric ES (F002), **NGS gene panels (>370 IEI genes)**, **whole-exome sequencing**, and **increasingly whole-genome sequencing** are recommended, especially in children, early-onset, syndromic, or treatment-refractory cases. **Chromosomal microarray/FISH** for 22q11.2 deletion (partial DiGeorge). Targeted single-gene testing when a specific IEI is suspected. Adult isolated ES has lower monogenic yield.

Clinical criteria & differential diagnosis. Diagnosis is clinical + laboratory (coexistent AIHA and ITP with positive DAT) after exclusion of: **thrombotic thrombocytopenic purpura / thrombotic microangiopathy** (check ADAMTS13; <10 IU/dL indicates iTTP), SLE, drug-induced

cytopenias, lymphoproliferative disease, DIC, and hemophagocytic syndrome. Microangiopathic hemolysis with schistocytes distinguishes TMA from the warm-antibody hemolysis of ES.

Screening. No population screening. **Cascade genetic screening** of relatives is appropriate when a monogenic IEI is identified. No newborn screening for ES.

11. Outcome / Prognosis

Survival/mortality (F003, F006). Adult median survival **7.2 years** (primary 10.9 years; **secondary only 1.7 years**, 5-year survival 38%). Pediatric survival at 15 years **84%** (~16% mortality), death at median age 18 years; pediatric HR for death 22× the general population. Leading causes of death: **infection, bleeding, and hematological cancer**.

Morbidity/function. High — chronic relapses, transfusion dependence during flares, cumulative multisystem immunopathology, and immunosuppression-related complications. Disease-specific QoL instruments were not identified (gap).

Complications. Recurrent/severe infections (major driver of mortality), thrombosis (including reports of Buerger's disease and spontaneous echocardiographic contrast), hematological malignancy, and progression to SLE.

Prognostic factors. Independent predictors of mortality: **number of second-line treatments** and **severe/recurrent infections** (F003). **Secondary etiology** predicts markedly worse survival (F006). ANA positivity + age >10 + high titer predicts SLE progression (F005). Immunopathological manifestations reduce splenectomy benefit.

12. Treatment

Line	Intervention	Evidence / notes	NCIT
First	Corticosteroids (prednisone/ prednisolone) ± IVIG	Standard; relapse common on taper (F004)	NCIT:C305 (steroid); NCIT:C555 (IVIG)
Second	Rituximab (anti- CD20)	75% response, 72% steroid withdrawal, 6-yr RFS 48% (F004)	NCIT:C1702
Second	Sirolimus (mTOR inhibitor)	Durable CR in refractory multilineage AIC, all ALPS children (F004)	NCIT:C1212
Second/other	Mycophenolate mofetil, azathioprine	Effective in ES + autoimmune hepatitis case reports	NCIT:C2005 / NCIT:C264
Targeted	Abatacept (CTLA4-Ig)	For CTLA4/LRBA defects	NCIT:C65483
Targeted	Leniolisib / sirolimus	For APDS/ALPS (PI3K-δ pathway); leniolisib validated in murine ALPS	—
Targeted	Ruxolitinib / baricitinib (JAK inhibitors)	Immune dysregulation subtypes	—
Complement/ other	Iptacopan (factor B inhibitor)	Refractory C3d-positive AIHA flares (case reports)	—
Surgical	Splenectomy	Effective but benefit reduced by immunopathological manifestations; infection/thrombosis risk	NCIT:C51915
Cellular	Hematopoietic stem cell transplant	For severe monogenic IEI-associated ES	NCIT:C15431

Pharmacogenomics/personalized medicine. Genotype-directed therapy is a defining trend: abatacept for CTLA4/LRBA, sirolimus/leniolisib for ALPS/APDS, JAK inhibitors for interferon/JAK-STAT–driven dysregulation. **Treatment strategy:** escalate from steroids/IVIG → rituximab/sirolimus → genotype-targeted agents; reserve splenectomy for selected cases; support with transfusion, infection prophylaxis, and thrombosis awareness.

13. Prevention

Primary prevention: none (no modifiable cause). **Secondary prevention:** early recognition of coexistent cytopenias and prompt immunosuppression; **early IEI genetic diagnosis** enables targeted therapy and family counseling. **Tertiary prevention:** infection prophylaxis (vaccination, especially before/after splenectomy — encapsulated-organism vaccines; antibiotic prophylaxis post-splenectomy), thrombosis vigilance, monitoring for SLE progression and malignancy, and surveillance for accruing immunopathological manifestations. **Genetic counseling** is indicated when a monogenic IEI is found (variable inheritance patterns). No immunization prevents ES itself.

14. Other Species / Natural Disease

Taxonomy. Naturally occurring ES is well documented in the **domestic dog** (*Canis lupus familiaris*, NCBI:txid9615) — concurrent immune-mediated hemolytic anemia and immune-mediated thrombocytopenia (F009). Reported breeds: **Rottweiler, Miniature Schnauzer, Dachshund**. Canine cases are DAT-positive and glucocorticoid/immunosuppressant-responsive, and can be complicated by thrombosis and opportunistic infection during immunosuppression — closely paralleling human disease.

Orthologous genes. *FAS*, *FASLG*, *CTLA4*, *LRBA*, *PIK3CD* orthologs are conserved across mammals. **Comparative biology:** the FAS apoptosis mechanism is evolutionarily conserved, underpinning both canine natural disease and rodent models. **Zoonotic potential:** none (autoimmune, non-transmissible).

15. Model Organisms

Model type. Mammalian (mouse) genetic models of the FAS pathway recapitulate ALPS-type ES (F009). **Specific systems:** **MRL/lpr (*Fas*^{lpr})** mouse (*Fas* loss of function) and **gld (*Faslg*)** mouse (*FAS* ligand defect) — both develop **lymphoproliferation, TCRαβ⁺ DNT-cell accumulation, autoantibodies, and autoimmune cytopenias**.

Applications. These models study the apoptosis-defect mechanism, DNT-cell biology, and therapeutics — e.g., **leniolisib reduced lymphoproliferative disease in murine ALPS** (F009), validating targeted therapy translation. **Limitations:** *lpr/gld* mice best model the ALPS/FAS subtype and background-dependent lupus-like autoimmunity; they do not capture the full heterogeneity of human ES (CTLA4, LRBA, APDS, NFKB1, SASH3, secondary ES). **Resources:** MGI, IMSR for Fas/Faslg alleles.

Mechanistic Model / Interpretation

Evans syndrome should be conceptualized as a **shared downstream phenotype produced by many upstream lesions of immune tolerance**. The unifying "trunk" is autoantibody-mediated, Fc-receptor/complement-driven splenic destruction of red cells and platelets. The "roots" are diverse: defective apoptosis (ALPS/FAS pathway), failed Treg checkpoints (CTLA4/LRBA), signaling hyperactivation (PIK3CD/APDS), transcriptional haploinsufficiency (NFKB1), adaptor loss (SASH3), or a structural syndrome (22q11.2/pDGS) — and, in a large fraction, no identifiable single gene (polygenic/multifactorial autoimmunity), or a secondary driver (SLE, lymphoma).

This model explains the clinical behavior: because the root cause is systemic immune dysregulation, ES is rarely "just" a blood disease — it tends to recruit additional autoimmune and lymphoproliferative manifestations over time (74% by age 20), progresses to SLE in high-risk ANA-positive children, relapses despite treatment, and kills primarily through infection (reflecting the immunodeficiency side of the dysregulation) and bleeding. It also explains why therapy is migrating from broad immunosuppression toward **mechanism-matched targeted agents** (abatacept, sirolimus/leniolisib, JAK inhibitors), which requires molecular diagnosis via genomic sequencing.

Evidence Base

PMID	Study	Supports
31983745	ES case report/review	Definition, autoantibody mechanism (F001, F008)
21228033	French AIHA cohort (n=265)	37% ES frequency; 74% IgG/IgG+C3d DAT (F001, F008)
41560547	Prospective AIC biomarker study (n=104)	51% IEI, 26% monogenic; pDGS/NFKB1/CTLA4/FAS (F002)
37792884	Adult ITP/Evans IEI screen (n=44)	Low adult monogenic yield (F002)
37646304	SASH3 case	Novel monogenic cause (F002)
33440924	OBS'CEREVANCE long-term (n=151)	Remission rates, 84% 15-yr survival, mortality factors (F003)
26484337	OBS'CEREVANCE cohort (n=156)	High treatment burden (F003)
28444729	Rituximab pediatric cohort (n=61)	75% response, 72% steroid withdrawal (F004)
26504182	Sirolimus prospective trial (n=30)	Sirolimus efficacy in refractory AIC (F004)
38227934	ANA-associated AIC study	45% ES→SLE; risk factors (F005)
31292991	Danish adult registry (n=242)	Incidence/prevalence; primary vs secondary survival (F006)
32271826	Danish pediatric cohort (n=21)	Pediatric incidence; 22× mortality HR (F006)
22157362	ALPS review	FAS apoptosis mechanism (F007)
38700373	ALPS biomarker study	sFASL/DNT biomarkers (F007)
31432443	LRBA defect series	LRBA tolerance mechanism; 93.3% splenomegaly (F007, F008)
19411652	Canine ES case	Natural disease in dog (F009)
41608120	Murine ALPS + leniolisib	Mouse model + targeted therapy (F009)
35443028	Splenectomy outcomes	Immunopathological manifestations reduce splenectomy benefit
40809448	Microangiopathic anemia review	TTP/TMA differential (ADAMTS13)

Limitations and Knowledge Gaps

- **No dedicated OMIM/ICD entry:** ES is coded by its components; identifiers are inherited from underlying IEIs, complicating standardized annotation.
- **Adult vs pediatric divergence:** monogenic IEI yield is high in children but low in adults; conclusions from pediatric cohorts may not transfer to adult-onset disease.
- **QoL data absent:** no ES-specific EQ-5D/SF-36/PROMIS evidence was identified.
- **Epigenetics unstudied:** no ES-specific methylation/histone or single-cell/spatial multi-omics signature was found in the reviewed literature.
- **Primary/idiopathic ES mechanism inferred:** for the large fraction lacking an identified gene, the tolerance-defect chain is extrapolated from monogenic cases, not directly demonstrated.
- **Two citation snippets were flagged "mismatch"** in the knowledge state

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— title-based); their claims (canine ES; murine ALPS + leniolisib) are corroborated by companion verified citations but should be re-verified against full text.

- **Treatment evidence** is largely from cohorts, single-arm trials, and case reports; randomized comparative data are scarce given rarity.

Proposed Follow-up Experiments / Actions

1. **Systematic genomic testing study** stratified by age of onset to define IEI yield thresholds and cost-effectiveness of WES/WGS in adult vs pediatric ES.
2. **Prospective biomarker panel** (DNT cells, sFASL, ANA titer, IL-10, immunoglobulins) to build a validated risk-stratification model for SLE progression, malignancy, and mortality.
3. **Genotype-stratified therapeutic trials** matching targeted agents to mechanism (abatacept for CTLA4/LRBA; leniolisib for APDS; JAK inhibitors for interferonopathy-like dysregulation).
4. **Single-cell and spatial immune profiling** of spleen/marrow/blood to map the autoreactive B-cell and DNT-cell compartments and identify novel targets.

5. **ES-specific QoL/PRO instrument development** and longitudinal capture within existing registries (OBS'CEREVANCE, Danish).
 6. **Comparative canine studies** leveraging naturally occurring canine ES as a translational large-animal model for therapeutics.
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Report compiled from 9 confirmed findings and 34 reviewed papers across 5 investigation iterations. Evidence source types: predominantly human clinical (national cohorts, registries, case series), with model-organism (Fas^{lpr/gld} mouse) and natural-disease (canine) corroboration.

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